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Score: _____

Please answer the questions below → Select the correct answer(s).
More than one answer may be possible.
A score of 20 correct answers is required to pass for this questionnaire.

1. **Entry to the EKL labs together with another person using one badge card is:**
 - ☐ Allowed if both users have at least one year experience.
 - ☐ Allowed if the badge owner agrees with it.
 - ☐ Allowed if you forgot or lost your badge.
 - ☐ Not allowed.
2. **Which statements are correct about the EKL buddy system?**
 - ☐ When working in wet benches or fume hoods a buddy is always required.
 - ☐ A buddy is only required if you use the wet benches & fume hoods outside the standard working hours.
 - ☐ Each lab user is responsible for finding his/her own buddy.
 - ☐ Lab buddies must contact each other at least every 15 minutes, to check if everything is OK.
 - ☐ Buddies may not change during lab use.
3. **Safety glasses must be worn:**
 - ☐ If you don't have your own glasses.
 - ☐ Only if you work with strong acids and bases.
 - ☐ When using the wet benches & fume hoods.
4. **Select the cleanroom rule(s) that are TRUE.**
 - ☐ Drinking or eating in the cleanroom is allowed if you are consecutively working longer than 3 hours.
 - ☐ Shorts and sandals are allowed in the cleanroom, but only during the summer.
 - ☐ Users must always clean up their working area before leaving.
 - ☐ A user is not allowed to use more than three coloured process wafer boxes.
 - ☐ Users must use coloured boxes and dedicated tweezers when working with "RED" metals.
 - ☐ Once you are dressed in a cleanroom suit and have entered the cleanroom, you are not allowed to unzip the suit.
5. **Users are allowed to use any lab equipment:**
 - ☐ After getting access to the EKL labs.
 - ☐ After getting permission from your promotor or mentor.
 - ☐ After an equipment training that was performed by an experienced colleague.
 - ☐ After completing the Safety & Introduction training course and the required equipment training by the tool owner or an allowed tool trainer.

6. You ask for equipment training via the:

- EKL website.
- EKL SharePoint site.
- NanoLabNL Information System (NIS).

7. Which statements are correct about the use of EKL equipment?

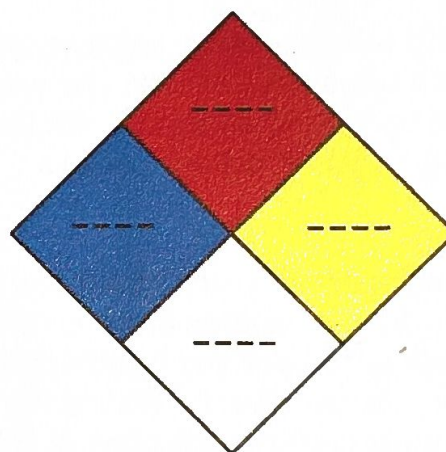
- It is not allowed to use equipment which is under maintenance and is marked with a STOP sign.
- You are allowed to repair or take corrective actions on the tool if you have used the equipment for more than one year.
- It is allowed to change the default system settings on equipment if you are an allowed user on it.
- If you need to use equipment for an extended period of time (>6 hours), then for ethical reasons it is recommended to start your process after 5 PM and let it run overnight.
- Any use or operation of equipment that differs from the procedure described in the operating manual must first be approved by the tool owner.

8. Which of the following statements about chemicals are true?

- MSDS stands for Material Safety Data Sheet.
- Ordering and using new chemicals in EKL that were never used before must be approved by the Wet Bench team.
- You are allowed to mix chemicals in the cleaning and etching lines.
- It is allowed to store chemicals in the fume hoods.

9. Put the corresponding meaning on the following MSDS picture:

- A** - Specific hazard information.
- B** - Reactivity information.
- C** - Health hazard information.
- D** - Fire hazard information.



10. What do you need if your process involves the use of a "RED" metal

- Approval from the process manager is sufficient.
- You need approval from the engineer that is responsible for the equipment that you want to use.
- A process flowchart that complies with the EKL PCC document and Process Guide Table, and approval from the contamination officer.
- A special process wafer box and dedicated tweezers.

11. Which documents are required to create a flow chart in which a "RED" metal will be used?

- The Process Guide Table (PGT).
- The EKL Safety & Behaviour Rules document.
- The Preventive Cross Contamination document (PCC).
- The MSDS of the "RED" metal.

12. Written flowcharts need to be in the EKL format. The template flowchart can be very useful. Where can you find it?

- On the EKL web site.
- On the EKL SharePoint site.
- On the NanoLabNL Information System (NIS).

13. How long is the Safety and Introduction training valid?

- As long as your access permit is valid.
- For MSc students 9 months, for PhD students 4 years.
- Only for one year. After that period you must successfully complete a refreshment questionnaire. You will receive a message from the NanoLabNL Information System (NIS).
- Until you deliberately violate the safety rules.

14. Which of the following statements are true?

- If I need wafers for processing, then I have to contact the process manager.
- Every person who is working after 18:00 in the CR100 is considered to be the last person and has to perform a lab check before leaving.
- Every CR100 user gets a starter kit, consisting of safety glasses, tweezers, a timer, a cleanroom notebook and a cupboard. The use of normal paper is not allowed.
- The Engineer On Duty (EOD) is responsible for repairing a tool and cleaning of the wet benches or fume hoods.
- It is not allowed to perform any processing in EKL without an approved flowchart.

15. Which of the following statements are correct?

- Only the most severe incidents or accidents need to be reported.
- Press the appropriate emergency button in case of an accident in any lab.
- The only persons responsible for the safety in the EKL labs are the director and the safety officer.
- In a case of a chemical burn of the eye, forcibly hold it open to ensure that the entire eye is flushed with an appropriate treatment liquid.

16. When do you have to press the calamity button?

- In case of a chemical spill larger than a few drops.
- If someone is exposed to a chemical liquid.
- If somebody dropped empty laboratory glassware.
- If a machine is leaking cooling water.
- If someone is exposed to a chemical fume.

17. What do you do when the fire or gas alarm sounds?

- You try to figure out why the alarm has gone off.
- You finish your experiment if you only need a little bit of time, turn your equipment off and exit the building.
- You take off your cleanroom suit and exit the building.
- You warn other users in your vicinity of the need to evacuate the lab, and exit the lab immediately.
- You check the lab to see if somebody needs help, and then leave the lab.

18. When do you press an equipment EMO button?

- If you suspect that a machine is not working properly and poses a threat to the lab or people in the lab.
- When a fire is noticed in any space of the cleanroom.
- When cooling water is leaking out of the machine.
- If you started a wrong recipe.
- If smoke is coming out of the machine.

19. What do you do when you contaminate a tool?

- You call the EKL facility team.
- You inform all other lab users.
- You turn the label on the tool to 'STOP', change the tool status in the NanoLabNL Information System to 'down', and inform the contamination officer.
- You leave the lab, and go for a coffee break.
- If the machine does not generate an error message, you finish your process step and continue working.

20. What do you do when you spot or cause a fire in any of the EKL labs?

- You panic and run home.
- You try to save your notes and rescue your work.
- You call the fire department.
- You start collecting all the fire extinguishers that you can find for the Emergency Response Team (ERT), so they can use them when they arrive.
- You press the fire alarm button and leave the lab.
- You take your phone and start making pictures.
- You go to the evacuation point and wait for the Emergency Response Team (ERT) and/or fire department to arrive.

21. What is important for your safety as a user from any EKL lab?

- You must know and follow the safety rules.
- You have to know who the lab manager is.
- You need to know how many fire extinguishers are available in the lab.
- You must know where the emergency exits are.
- You should be able to find the emergency showers and safety kits in the lab.
- You must always have a phone with you, so you can call for help.

22. What is correct?

- Users are allowed to clean up minor chemical spills (a few drops) by themselves.
- Fire extinguishers can only be used for fires smaller than a paper basket.
- The eye wash bottles are located nearby the safety showers and the wet benches.
- It is not allowed to scratch your face with your gloves on.

23. What does the Dutch message "Gas alarm, cleanroom verlaten" mean in English:

- Gas alarm, clean room is not safe.
- Gas alarm, dangerous.
- Gas alarm, leave the cleanroom.

24. Where are the EKL evacuation points located?

- In front of the measurement labs.
- In the office of the EKL safety officer.
- In the EKL reception hall at the main entrance (in front of office EKL00.010).
- In the main hall of the EWI building.
- On the field in front of EKL's main entrance and on the Mekelpark (in-between the EWI and CITG building).